Test Objectives

Every case in `forensic-agent-tests` should be traceable to one or more of the nine forensic-capability categories below. This document defines each category and tracks candidate sources — existing datasets, open-source tools, and relevant procedural skills — that could inform or supply a case covering it. It's a planning reference, not a coverage guarantee: a source appearing here means it's worth evaluating for a category, not that it's already been vetted or adopted.

A note on the "Anthropic-Cybersecurity-Skills" library

The Related Skills column below references `Anthropic-Cybersecurity-Skills` (local checkout: `/Users/mlcs/Documents/github/Anthropic-Cybersecurity-Skills`), 818 procedural cybersecurity skills mapped to MITRE ATT&CK/NIST CSF and other frameworks. **Despite the name, its own README states it is an independent community project, not affiliated with Anthropic PBC** — cite it as such wherever it's referenced.

Two distinct ways it's used, and both are now decided:

1. **Case-design input**: mine these skills' descriptions, tool references, and MITRE ATT&CK mappings when authoring `EXAM.md` questions and `grading_schema.md` rubrics, so cases reflect what a real analyst's workflow actually checks for a given category. This applies regardless of what the AUT has access to — it makes the *case* more realistic, not just the AUT's toolkit.

2. **AUT toolkit — adopted.** The agent-under-test is enriched with this skills library during investigation. This does shift what's measured toward playbook-following as well as native reasoning, and that's intentional. It does not preclude running the same cases against non-enriched models (Anthropic or otherwise) for comparison — the cases themselves stay valid either way, since case-design input (above) never depended on AUT enrichment in the first place.

   Delivery mechanism (noted 2026-09-09, see `../../EVAL_SUITE_PLAN.md`): this decision was adopted without a concrete way to hand the skills to an AUT at run time. There now is one, contingent on the harness's still-open framework choice — EvalScope's `NativeAgentConfig(skills_dir=...)` reads host folders containing `SKILL.md` natively, no scaffolding required. Two things this raises that belong to *this* document rather than the harness plan:

   - **Which skills get mounted is a test-design question, not a config detail.** Mounting a category-matched subset would tell the AUT which of the nine categories below the case belongs to — a hint it isn't supposed to have, and one that partly answers the question the case is asking. Mounting all 818 avoids the leak but changes what's measured (skill *selection* becomes part of the task). **Decided 2026-09-18: the same 10 skills for every case**, the union of `TEST_CASE_MATRIX.md`'s "Skills Enabled" column. Each case's own list is written with knowledge of its answer, so mounting it per case would hint at the finding; the shared union reveals nothing about any one case. All 818 was rejected on prompt size. Record the library commit and the 10 names alongside any published score.
   - **An enriched run and a non-enriched run are different experiments.** The paragraph above already says both are valid; make sure the enrichment state is reported with the result, since a score is not comparable across that boundary.

   Vendoring (submodule vs. directory vs. pinned copy) remains open — see `TODO.md` item 3. Given the library is an independent community project (see the caveat above), a pinned copy is the safer default: an upstream change to a skill silently changes what the benchmark measures.

Other skill repos surveyed (not yet adopted for AUT enrichment)

Same "independent, not Anthropic-affiliated" pattern holds for both:

- `briiirussell/cybersecurity-skills` — 33 skills, MIT. Most relevant: `disk-forensics` (evidence recovery, timeline reconstruction), `incident-triage` (NIST SP 800-61), `security-comms` (audience-tailored incident comms/post-mortem/breach-disclosure templates — directly useful for category 9 rubric design, arguably richer than anything found for that category in Anthropic-Cybersecurity-Skills).
- `transilienceai/communitytools` — 27 skills, offense-oriented (pentesting/bug-bounty lifecycle). Includes a `/dfir` skill (Windows event logs, PCAP analysis, AD attack detection, executive+technical report generation). Its main value here

isn't defensive analysis — it's realistic *attacker*-side tradecraft, useful for keeping EvidenceForge storyline authoring grounded in what a real operator's exploitation chain looks like, independent of any AUT-facing skill.
- The `agentskills.io` site (cited by Anthropic-Cybersecurity-Skills as "the standard" it follows) is the Agent Skills *format spec*, not a registry of skill libraries — it lists client applications that support the format, not skill collections. Dead end for finding more libraries this way.
- Claude Code's own built-in `security-review` skill is a first-party option distinct from all of the above community libraries, worth keeping in mind separately since it isn't a third-party dependency.

Categories

1. Log Analysis

Ability to read and reason over raw Windows and Linux log sources (Security and Sysmon events, syslog, application logs), search for specific keywords, identifiers, or indicators across a volume of records, and extract or filter by time range to isolate relevant activity.

(!) 2. Browser History / Download History Verification

Ability to review browser artifacts — history, downloads, cache, typically from Chrome or Firefox — to reconstruct what a user searched for, visited, or downloaded, and correlate that activity against a broader investigation.

Known deficiency, tracked not blocking: no current tool in our generation pipeline (EvidenceForge or otherwise) produces browser artifacts. NIST CFReDS Data Leakage Case remains the identified path to closing this (see the Coverage Matrix and `TEST_CASE_MATRIX.md`), but it requires disk-image case infrastructure we haven't built yet. This category has zero case coverage until that's done — not a gap to paper over with a weaker substitute.

3. Account Activity Verification

Ability to analyze authentication and session evidence (Windows Security auth events, Linux auth logs, SSH/sudo activity) to establish who authenticated, from where, when, and whether that activity is consistent with a single legitimate identity.

4. Network Connection History Verification

Ability to analyze host connection and traffic evidence — firewall, proxy, DNS, VPN, netflow/connection-record logs — searching by IP address or time range to reconstruct what a host connected to, from where, and whether any of it warrants scrutiny.

5. Timeline Reconstruction

Ability to assemble a coherent, correctly-ordered sequence of events from multiple, independently-timestamped sources into a single chronological narrative.

6. Intrusion Path Identification

Ability to reconstruct how an attacker gained a foothold — the specific technique, vulnerability, or vector exploited — from the available evidence, without being told the vulnerability in advance.

7. Lateral Movement Analysis

Ability to trace movement from one host or account to another — an attacker pivoting, or in a benign case, a legitimate user working across systems — correlating evidence across the source and destination systems.

8. Data Exfiltration Indicator Analysis

Ability to recognize evidence consistent with data leaving an environment without authorization — anomalous transfer volumes, timing, or channel (e.g. DNS tunneling, encoded beaconing) — and assess what, if anything, was actually exfiltrated.

9. Report Generation

Ability to synthesize findings from an investigation into a clear, accurate, appropriately-hedged report for a given audience, distinguishing confirmed facts from working hypotheses and unconfirmed details.

Coverage Matrix

Index #	Category Name	Example Scenario	Possible Open Source Tools / Sources	Related Skills (Anthropic-Cybersecurity-Skills)
1	Log Analysis	A week of Windows Security + Linux syslog exports from a mixed-OS network; locate every event referencing a given process name or username within a bounded time window.	EvidenceForge (Windows Security/Sysmon XML, syslog, bash_history); AIT Log Data Set v2.0 (real multi-service Linux logs); EVTX-ATTACK-SAMPLES / EVTX-to-MITRE-Attack (real binary EVTX for parser validation); OTRF Security-Datasets (Mordor) / splunk/attack_data; `plaso`/`log2timeline` for cross-format parsing checks	`performing-log-analysis-for-forensic-investigation`, `performing-linux-log-forensics-investigation`, `extracting-windows-event-logs-artifacts`, `analyzing-windows-event-logs-in-splunk`, `generating-forensic-timelines-with-hayabusa`
2	Browser History / Download History Verification	An employee under suspicion of pre-resignation data theft; review their workstation's Chrome/Firefox history, downloads, and cache to reconstruct what they researched and downloaded, and when.	NIST CFReDS Data Leakage Case (disk image + removable media, analyst-level answer key); `browser-history` (Python extraction library); Hindsight (Chrome/Chromium forensic timeline tool); Playwright/Selenium (drive a real browser to generate genuine synthetic artifacts for a scripted persona)	`analyzing-browser-forensics-with-hindsight`, `extracting-browser-history-artifacts`
3	Account Activity Verification	A shared server shows two concurrent SSH sessions authenticating as the same user from two different source IPs; determine whether this is a physically impossible/shared-credential pattern.	EvidenceForge (4624/4625/4634/4648/4672/4720-4757/4768-4776, sshd/sudo/su); AIT Log Data Set v2.0 (real auth.log); OTRF Security-Datasets (credential-access-technique-mapped captures)	`analyzing-linux-audit-logs-for-intrusion`, `performing-active-directory-compromise-investigation`, `detecting-ntlm-relay-with-event-correlation`, `detecting-golden-ticket-attacks-in-kerberos-logs`, `detecting-service-account-abuse`, `detecting-email-account-compromise`
4	Network Connection History Verification	A firewall/proxy/DNS log bundle covering a DMZ segment; identify all external connections to/from a given host within a time range and flag anomalies.	EvidenceForge (Zeek conn/dns/http/ssl/files, Cisco ASA, proxy, Snort/Suricata); AIT Log Data Set v2.0 (VPN, firewall, Suricata, PCAPs); DARPA OpTC (500-host scale); LANL Unified Host and Network Data Set (real enterprise data, benign substrate); Zeek / Wireshark / `tshark` for PCAP reprocessing	`performing-network-forensics-with-wireshark`, `performing-network-packet-capture-analysis`, `analyzing-network-flow-data-with-netflow`, `performing-network-traffic-analysis-with-zeek`, `detecting-network-anomalies-with-zeek`, `implementing-network-traffic-baselining`

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5	Timeline Reconstruction	Multi-source logs spanning several hours across hosts; reconstruct the exact chronological sequence of a user's or attacker's actions.	EvidenceForge (cross-source consistent timestamps); OTRF Security-Datasets (Mordor); `plaso`/`log2timeline` + `psort`; Timesketch	`building-super-timelines-with-plaso`, `performing-timeline-reconstruction-with-plaso`, `generating-forensic-timelines-with-hayabusa`, `building-incident-timeline-with-timesketch`; also `briiirussell/cybersecurity-skills`' `disk-forensics` (evidence recovery + timeline reconstruction)
6	Intrusion Path Identification	A public-facing web server is compromised; reconstruct the exploitation chain from initial recon through code execution using web, network, and endpoint evidence.	EvidenceForge (`web_scan` storylines, IDS layering); AIT Log Data Set v2.0 (labeled attack steps); OTRF Security-Datasets / splunk/attack_data (ATT&CK-mapped intrusions); EVTX-ATTACK-SAMPLES	`analyzing-cyber-kill-chain`, `performing-active-directory-compromise-investigation`, `analyzing-indicators-of-compromise`, `collecting-indicators-of-compromise`, `triaging-security-incident-with-ir-playbook`; also `transilienceai/communitytools`' `/dfir` and offense-side skills — useful for realistic attacker tradecraft when *authoring* the storyline, not for the AUT
7	Lateral Movement Analysis	After an initial workstation compromise, the attacker pivots to a second internal host via RDP/PsExec; trace the pivot using correlated logon and network evidence across both hosts.	EvidenceForge (4648 source + 4624 target correlation, RDP, cross-host RSAT sessions); DARPA OpTC (large-scale lateral movement, needle-in-haystack); OTRF Security-Datasets (PsExec/WMI/RDP-mapped captures)	`detecting-lateral-movement-in-network`, `detecting-lateral-movement-with-zeek`, `detecting-lateral-movement-with-splunk`, `hunting-for-dcom-lateral-movement`, `hunting-for-lateral-movement-via-wmi`, `performing-lateral-movement-detection`
8	Data Exfiltration Indicator Analysis	A compromised host establishes a slow, low-volume DNS-tunneled channel to an external domain over several hours; recognize the volumetric/timing anomaly and assess what may have left the network.	EvidenceForge (`dns_tunnel`, `spillage`, `beacon` byte-growth — reverify exact current coverage); AIT Log Data Set v2.0 (`dnsteal`-labeled DNS exfiltration); DARPA OpTC	`hunting-for-data-exfiltration-indicators`, `hunting-for-data-staging-before-exfiltration`, `detecting-dns-exfiltration-with-dns-query-analysis`, `detecting-exfiltration-over-dns-with-zeek`, `hunting-for-dns-tunneling-with-zeek`
9	Report Generation	Given a fully investigated case's raw findings,	NIST CFReDS (the only surveyed source with analyst-conclusion-level ground truth, not just line-level	`building-incident-response-playbook`, `building-malware-incident-communication-`

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		produce a concise incident report for non-technical stakeholders that correctly separates confirmed facts from unconfirmed hypotheses.	labels); DFIR-Metric (arXiv 2505.19973) — prior-art LLM DFIR benchmark methodology, not a data source	`template`, `conducting-post-incident-lessons-learned`, `generating-threat-intelligence-reports`; also `briiirussell/cybersecurity-skills`' `security-comms` (seven audience types, incident comms/post-mortem/breach-disclosure templates — the richest report-generation source found so far)